

Distributing operational support to transform business operations

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Operational support is increasingly a key differentiator for telecommunications operators. It is an essential part of launching and delivering telecommunications services. However, the quickening pace of new product and service launches, together with increasing customer expectations and changes in commercial structures for service delivery, all highlight the need for more imaginative ways of providing operational support. This paper describes some innovative work at BT to introduce collaborative working into the end-to-end processes of operational support, between supplier and customer. It has applied the principles of community of interest networks (running over extranets) to back-office processing in the carrier business, between operators. Distributed systems principles and technologies underpin the technical design, and business benefits are available to all the organisations using it.

1. Introduction

Handling direct customer interactions (helpdesk, sales and marketing, fault handling, etc), and managing services and underlying networks, are essential parts of launching and running telecommunications services. The quality of these operational support processes and innovations to improve their effectiveness are increasingly key differentiators for telecommunications operators. Systems to provide these support functions have mainly been developed over the last 20 years. A good insight into BT's operational support systems (OSS) is given in Stockman [1], where the business processes involved, some of the technical challenges, and the solutions embodied in BT's systems are described.

OSS are typically regarded as back-end systems. It is traditional for customers to rely entirely on the telecommunications company or service provider to carry out the business processes and customers have little or no direct control over them. A recent development has been the beginnings of 'self-service' access for some well-defined functions, e.g. changing telephone numbers for bill discounts, with access via interactive voice recognition, automated speech dialogues, or World Wide Web. Operationally, however, these are still only extra access methods to a system run entirely by the telecommunications company.

The systems running these conventional operational support processes are usually large and based on centralised processing. Early adoption of centralised computing by telcos was pioneering in its innovation — and an evolution path can be traced from the early days of batch processing. Distributed systems technology is now gradually being

deployed — the back-end systems are enhanced to include client/server working [2], on-line transactions, and even some real-time interactions between different systems [3]. However, the philosophy of distributed systems (and, even more so, the philosophy of IP and the Internet) has not yet become accepted as a central design principle.

There are benefits, of course, from the centralised approach — easier security, greater control of the processes and access to them, the accumulated experience of designing, building and running such systems, and the mature run-time support infrastructure for them. However, there are also drawbacks, which are becoming more important in the context of increasing competition and rapidly changing markets. These can include long lead times (affecting the ability to launch new products), growth in stand-alone support systems (limiting the possibilities for product bundling and service interworking), low flexibility (leading to difficulty in making small changes in response to market conditions or marketing initiatives), and rapidly increasing complexity as more and more functions are added to existing centralised systems.

With the ever-quicken pace of new product launches, and particularly with the advent of on-line multimedia services, more imaginative ways of providing operational support are required. OSS need to be designed to have similar lead times to product and service developments.

This paper describes some innovative work at BT to introduce collaborative working into the end-to-end processes of operational support, between supplier and customer. It has applied the emerging technologies and

business principles of community-of-interest networks (CoINs) (running over extranets) to the back-office processing involved in the inter-operator carrier business. Distributed systems principles and technologies underpin the entire technical design, and business benefits are available to all the organisations using it. The paper explains the rationale for the design approach (section 2), describes how the current OSS can evolve towards it (section 3), and surveys the technologies available to implement it (section 4). Section 5 describes a technology trial which is currently under way, involving several UK operators — and gives an indication of some of the early results from the trial.

2. Dispersal and enterprise models

The work carried out in BT is targeted towards a far-reaching re-examination of what processes operational support needs to provide, and how those processes can be delivered in the emerging telecommunications market. One of the central principles of this work is ‘dispersing OSS’.

2.1 The dispersed OSS vision

The purest vision of dispersed OSS offers a radically new approach with increased flexibility and is based on two concepts:

- dispersal — the decentralisation of responsibility, business processes, information and systems,
- mass customisation — empowering customers to configure and tailor the complete service offered, including the way it does business with the service provider.

As depicted in Fig 1, this dispersed OSS vision is applicable to various customer segments, although the effect of it in each case may be quite different (Fig 2).

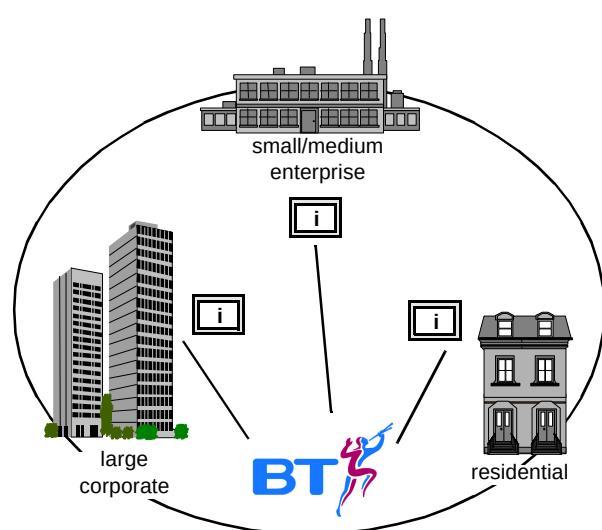


Fig 1 Dispersed OSS vision — integrated OSS with all customers based on dispersed information and processes.

Achieving the vision will build closer business relationships with customers. It is important to realise that the dispersed vision is not just about giving wider access to OSS, it is also about recognising that responsibility for some aspects of OSS logically belongs to customers, and that technology today allows these to be securely allocated more appropriately. Furthermore, it recognises that embodying the processes and data in only one organisation leads to extra work for everyone.

Allocating responsibilities appropriately removes unnecessary process steps and so reduces costs for everyone. At the same time it improves the quality of operational support both by eliminating whole classes of problems, and by requiring less operational support to be done within the service provider.

Dispersal is a continuous quantity, i.e. there are degrees of extent, and it can be said that some amount of dispersal is already implemented (e.g. the SIM card in a mobile phone disperses some responsibility, data and process to the user or customer, but in a limited way). A distributed architecture for OSS is the platform needed for fully dispersed operational support, but there are many opportunities to begin a dispersed approach with current OSS. Some of these opportunities have been advanced by the technology trial described later in this paper (section 5).

OSS functions include provisioning, fault reporting/management, and pricing/charging/billing. These are all potential examples of where dispersed, customisable OSS, which exploit distributed systems technology, could support better, more flexible operational support processes.

Where dispersal is employed across a number of OSS functions, the relevant ‘business objects’ can be regarded as co-operating entities, which interwork to provide the complete OSS capability required for a service and/or customer. A provisioning object, for example, can communicate with a billing object to ensure that any newly provisioned service can be billed in a consistent way for a particular customer.

A charge that is often levelled at the idea of dispersal is that, although some customer aspects (such as customisation and flexibility) are made easier, some corporate management aspects may be made more difficult (e.g. aggregation of operational statistics). In fact, this is just a case of design priorities. Just as one would not design a centralised system without non-functional requirements or error handling, one should not design dispersed systems without regard to house-keeping or management information gathering requirements.

Clearly the dispersal of OSS functions is not necessarily appropriate in all circumstances. It is another advantage of

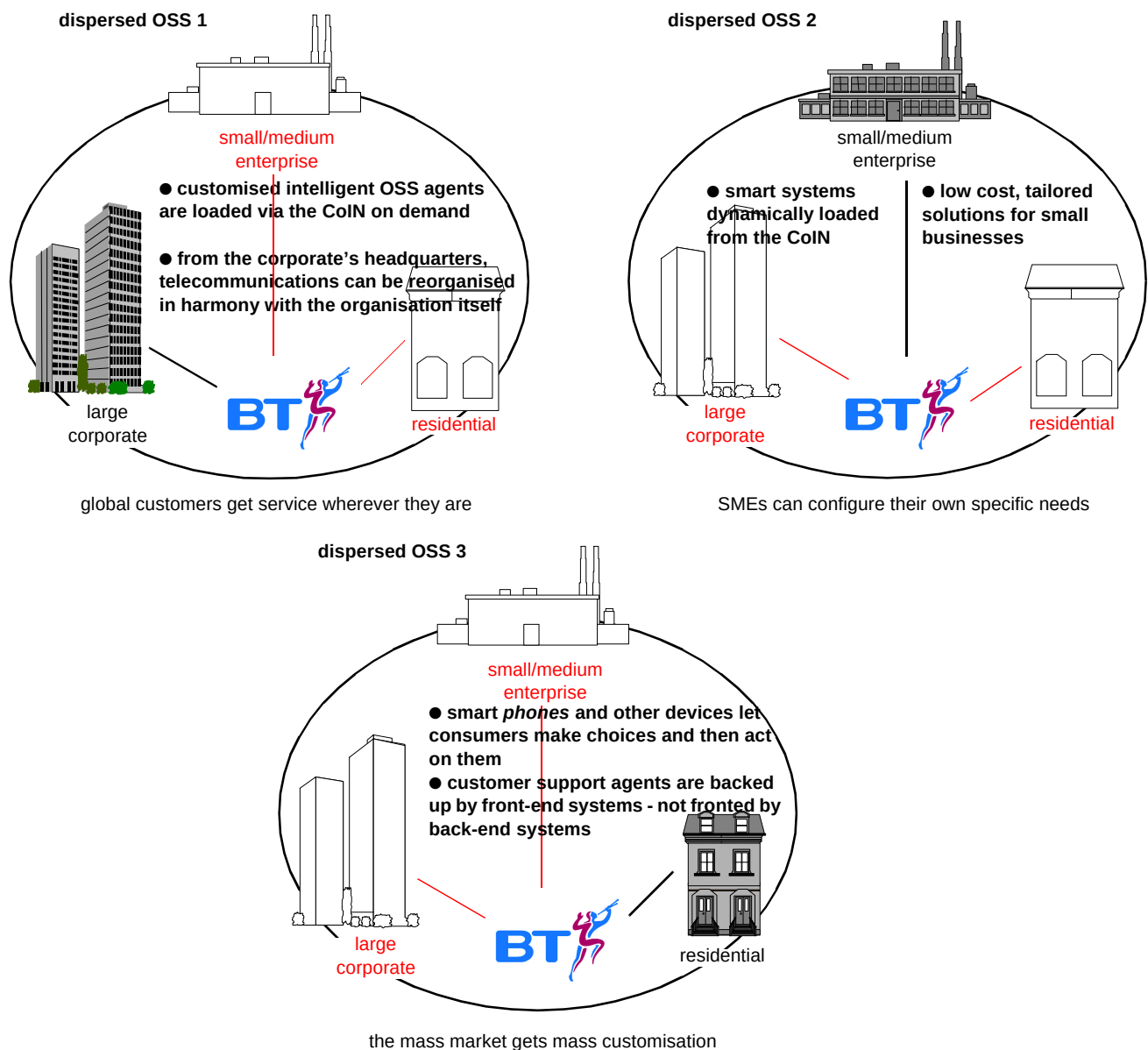


Fig 2 Alternative dispersed visions.

the approach, however, that it can be employed in a mix-and-match way with other approaches. Unlike some alternative visions, this is not an all-or-nothing idea. It can be targeted at specific instances where it may be most beneficial.

2.2 Enterprise modelling

The business model, underpinned by dispersed operational support, together with information sharing and collaborative applications, is a truly radical change compared to the customer/supplier type relationships that are commonplace today. For example, it requires sharing direct access to each other's business systems, dynamically and on-demand. The same radical contrast is evident when considering the internal supplier/consumer relationship that exists today between different business units in large companies like BT.

Enterprise modelling provides a formal means to reason about such changes. The enterprise modelling methodology followed in this work concentrates on the business responsibilities associated with the roles of the parties involved in providing service. The analysis and exploration of these roles and responsibilities can then lead to a more appropriate design for the systems that support an organisation's business goals. It also provides a stronger basis on which to decide the commercial arrangements for doing business in this revised way. The work described in this paper used enterprise modelling based on the methodology of Martin and Dobson [4].

3. OSS evolution

As with biological evolution, operational support can be regarded as passing through a number of distinct t

stages (Fig 3). The original stage was based on entirely manual systems. Later, automated centralised systems were developed, initially using batch processing, and then incorporating transaction processing. Interfaces between systems for different operations have since become commonplace leading to the three-tier architecture system structure. More recently, the trend has been to widen access via World Wide Web (WWW) interfaces and the Internet. One can imagine a later stage when many of the systems have been distributed. By this stage, not only will the technology allow parts of the system to run across various machines, but different OSS functions will be 'dispersed' and responsibilities will be transferred to others, e.g. customers, brokers, intermediate service providers. However, an intermediate stage, during which the newer Internet-based front-ends are combined through interfaces to existing back-end systems, can probably be expected. This will smooth the evolution process and allow the transfer of responsibility outside the service provider.

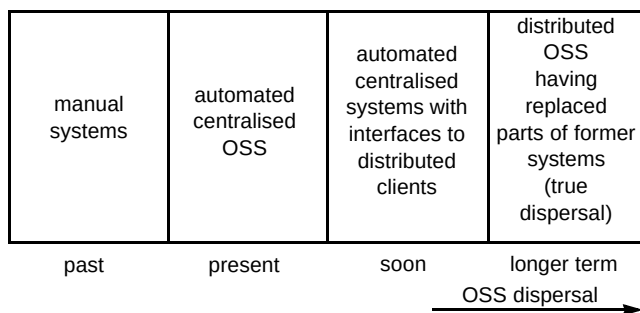


Fig 3 Staged evolution of OSS.

Just as the shift from manual to automated centralised systems resulted in major cultural and organisational change (e.g. employment changes, unforeseen functionality), so will transitions between future stages, though the nature of the changes involved will be different.

4. Bringing technologies together

From the technology viewpoint, achieving a dispersed OSS potentially utilises and combines:

- Internet technology, exploited as the basis for future OSS infrastructure through secure community-of-interest networks,
- collaborative working technology, from on-line conferencing to shared processes and support for inter-organisation workflow,
- distributed object technology, moving to smaller, managed clusters of distributed systems for billing, charging, provisioning, customer support, and fault management.

4.1 Extranet-based communities of interest

In the corporate world, it is becoming commonplace for organisations to base their internal processes on an intranet. An extranet capitalises on this by allowing secure links to be made between different organisation's intranets (Fig 4). The benefits of the intranet are then exploited between organisations doing business with each other. A CoIN is thus formed, over which organisations can share information and business processes.

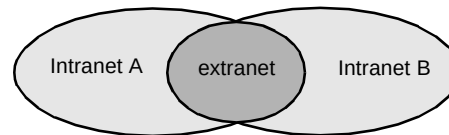


Fig 4 A secure extranet link between two intranets.

Although some extranets have been built and are in use by a few groups of companies, BT (due to its size) is pioneering the use of extranets on a scale not yet seen. Indeed this has been the case with its intranet experience as well. It is the first step in moving from large centralised OSS to more distributed, flexible and customer-focused systems. Because the number of successful examples of this approach is small and the technology relatively new, this evolution needs to be tried in appropriate niche domains initially in order to learn lessons for wider deployment. Customer demand for the facilities that are provided over this newer type of system should also play a significant role in driving this future direction of OSS. One such niche domain is the wholesale/carrier business, described later in this paper as a case study and technology trial.

Extending access to systems (by networking techniques) is nothing new. However, a number of the desirable aspects of the success of the Internet and WWW are preserved by the specific use of extranets.

Firstly, the non-proprietary nature of the network makes the interconnection of various organisations easier. In building a true partnership relationship, different organisations involved can use different technologies that interwork, including the core network and the client/server applications built on top. The open standards encouraged by the Internet provide an ideal basis for partnership interconnection. The extranet approach also inherently means that more benefit can be squeezed from existing intranet investments. This reuse of corporate infrastructure in each organisation is very desirable from a cost perspective.

Secondly, the sheer abundance of Internet/intranet/extranet interfaces in products both at the client front-end and the database back-end helps to de-risk extranet-based OSS with respect to future-proofing. Almost every new software product on the market now has some sort of Web

interface and, given the expansive evolution of the Internet, this can be expected to continue. Any system designed around this technology is very unlikely to become obsolete; rather it is likely to keep the organisations using it at the forefront of technology. Furthermore, as new products become available in future, an extranet infrastructure is likely to enable their smooth integration.

Finally, associated platform-independent technologies such as Java and network computing [5] make the development of truly distributed OSS across an extranet an easier proposition. The ability to quickly develop lightweight applications, supported by such technology, is ideally suited to a dispersed OSS. Software development tools aimed at providing some rigorous control over the development and deployment processes for distributed systems are now maturing [6]. The network infrastructure that enables the partnership interconnection will also facilitate the automatic deployment of software and its associated upgrades.

4.2 Collaborative working technology

The same extranet IP-based infrastructure used to build the CoIN can also be used to run distributed applications which support managed information sharing and associated responsibility for ownership as well as shared processes and workflow support. This can be extended to include applications for conferencing and collaborative working (Fig 5).

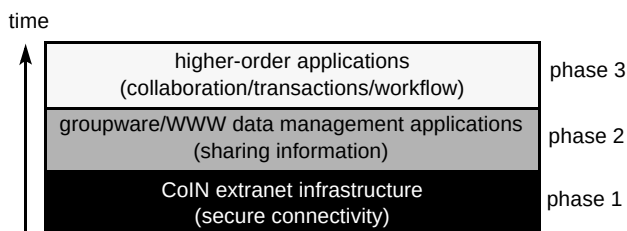


Fig 5 Extranet IP-based infrastructure extended usage.

4.3 Distributed object technology

Existing operational support systems are largely heavyweight monolithic centralised systems using older, more established technology. The operation and maintenance of these systems is a major business-critical activity for established telcos such as BT, since the ability to offer services and collect revenue is heavily reliant on these systems. To move from this to the reality of dispersed operational support will require their replacement by distributed OSS functions designed and implemented as distributed objects [7]. In the longer term, as mentioned above, this 'object' implementation will allow dynamic downloading of functions as needed to perform OSS functions over the same extranet IP-based CoIN infrastructure.

5. Case study and technology trial

5.1 Putting theory into practice

The vision presented provides a tantalising glimpse of how things could be in the not too distant future. Yet it will be apparent that this vision requires no specialist technology — that all the necessary technology needed to achieve the vision is already available. An early technology trial of essential parts of the vision was desirable. The UK carrier business was chosen for such a trial because of a number of important observations:

- it involves sophisticated back-office operations,
- commercial transactions for interconnect exhibit a 'partnership' style of interaction,
- there was a desire to automate a significant proportion of the carrier processes by most operators, including BT.

The primary element of the carrier business is interconnect, which occurs when telecommunications traffic (voice and data) passes from one operator network to another. However, this cannot be done on an *ad hoc* basis as each network has finite capacity. Hence, the key commodity of the carrier business is capacity. Operators have to forecast their likely requirements, while the carrier has to ensure that sufficient capacity exists within the carrier network to ensure that the operators' requirements are met.

The general vision, along with the idea for a technology trial, was presented to the Standards Contracts Forum, a forum hosted by BT which all UK operators are invited to attend. It was agreed that some form of evaluation should take place in order that the UK interconnect industry as a whole could ascertain the benefits. To this end, a technology trial was established during September 1998 and is expected to be completed at the end of May 1999. For various reasons, the trial has selectively implemented elements of the full vision and some aspects have been adopted in a less than optimal configuration. This still provides useful feedback. The primary aim of the trial was to test the general principles, including issues concerning the concept, culture and technology of a community-of-interest network. The operators who have been taking part, along with BT, are Cable and Wireless Communications (CWC), NTL, Orange and Scottish Telecom.

In summary, the aim of the trial is to evaluate the use of advanced concepts (such as dispersal) and technology in the interconnect industry. The enhancement of business relationships, using dispersed operating support functions, can be achieved in a number of ways. The optimal allocation of responsibilities is a key step — each organisation should be in control of its data and processes. This in turn brings about improved management control and a significant reduction in operating costs for all concerned

when used in conjunction with process automation; but it is improved inter-operator back-office processing which will really enhance the business relationship. While the linking of back-end systems on a routine basis is some way off, the trial has started the process in a small way.

5.2 System design

The trial is using a BT product, Intranet Complete [8], to build the CoIN, linking all the participants' intranets together. In addition, Microsoft Exchange was chosen as the groupware product [9], and Microsoft Outlook as the client software [10]. The choice of technology was primarily based on factors such as availability, cost, and the effort required to develop process application software (using Visual Basic).

The general configuration of the trial is given in Fig 6. It should be noted that the CarrierNet server is in fact a 'demilitarised zone' which lies outside the firewall of each triallist, effectively managed by a third party. The reasons for this are twofold. The first is that the trial system is symmetric — the processes in the trial are designed to be applicable to any operator supplying any other operator. Secondly, such a repository is the natural place for truly shared data, particularly in the early stages, when security issues prevent direct access between different operators' back-end systems. With regard to security, the CoIN has a firewall around the shared server. Partitioned data is stored on the server with password protection, with access authorisation controlled by the partition owners.

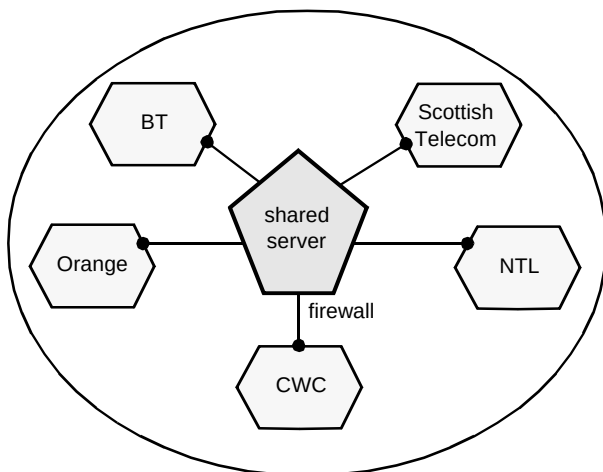


Fig 6 Trial configuration.

5.3 Applications and processes

Initial process applications developed for the trial have been related to the 'technical master plan' (TMP) — a set of files containing key information on the customer requirements (see Fig 7). Of particular interest have been those parts of the TMP which relate to pre-order forecasts and traffic forecasting. In the trial, the processes which use the TMP involve BT supplying capacity to other triallists. However, as stated earlier, the processes are designed to be symmetric.

Each process was chosen to exercise a particular part of the vision. The traffic forecast (TF) (an annual forecast of expected capacity requirements prepared by the operator) and the advanced capacity order (ACO) (a quarterly capacity forecast) are examples of supporting and automating shared processes between organisations. The customer service profile (CSP) (key contact information for both BT and the operator) represents an example of shared ownership and information between organisations. A TRIB table (a document summarising link and tributary information) is an example of a link, in this case indirect, between back-end systems — a key aspect of the trial.

The CSP is of particular interest as it contains information about both BT and the operator. Currently, as BT has responsibility for this document, any changes that need to be made regarding changes in operator details have to be made by BT. It would be far more sensible if the operator could change their details themselves. Hence, in the trial, this is a truly shared document — the operator has read and write access to its information, but read only access to BT information and vice versa.

The other elements of the trial involve process modelling and simple end-to-end workflow. To achieve this the system uses Microsoft Outlook and Exchange. Processes are automated using Visual Basic scripts over the Outlook directory structure. Much greater levels of sophistication could be achieved with a workflow tool, and to this end the trial is considering the inclusion of Oracle Workflow.

5.4 Trial metrics

Quantifying the benefits (or otherwise) of the trial is an important activity. Metrics are being collected to quantify

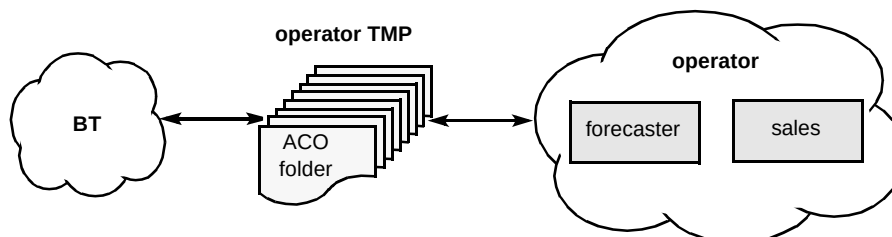


Fig 7 The technical master plan.

both the business and technical aspects of the trial. For example, the technical objectives of the trial are:

- sharing of data,
- better performance,
- reduced system administration,
- flexible and adaptive processes,
- process control,
- easy system evolution,
- secure access.

There is a similar set of objectives pertaining to the business goals of the trial. Using the goal-question-metric methodology [11] a set of metrics was devised in order to characterise the performance of the trial with respect to these criteria. On completion of the trial the data collected will be used for analysis in support of decisions for follow-on activities.

5.5 Results so far

Full results of the trial will not be known until the trial is completed, but some qualitative interim findings are already clear.

- E-business

It is clear that a CoIN infrastructure offers a good vehicle for the future generation of e-business in a dynamic interconnect market. However, as with the issue of changing relationships, the trial is highlighting a number of interesting issues which are, at present, not well understood, e.g. CoINs using the same network infrastructure. Indeed, the impact of CoIN technology in general is a significant research issue.

- A lever for change

The trial has acted as a catalyst in opening up the possibilities of new ways of working using dispersed system and data via an extranet.

- Changing relationships

The use of CoIN technology and process automation can radically alter the dynamics of the interworking between BT and its customers. The trial is highlighting the importance of well-defined processes and responsibilities in the changed way of working.

- Interfacing with the desktop

It is clear from the trial that a self-contained system is not adequate if it is separate from the normal desktop environment of the users. The CoIN must integrate directly into the corporate IT infrastructure and not to some stand-alone client(s).

- Heterogeneity

The degree of variability regarding IT infrastructure of the triallists is much higher than expected and significant diversity can therefore be anticipated in

other potential communities of interest. Consequently, the client software to access the system needs to be as ubiquitous as possible — probably just a Web browser.

- Process relationships

There is a strong desire among the triallists to link, or correlate, information from one process with another, for example, correlating actual orders against an advanced capacity order.

6. Conclusions

The work described in this paper is innovatively applying distributed systems, particularly the approach of community-of-interest networks, to back-office operational support functions between telecommunications organisations. Dispersal is an application of distributed systems and Internet principles in an area which has not traditionally been designed that way. It also extends the rationale for using distributed systems, by distributing data and functionality on the basis of business needs, not just on technical criteria like performance.

There are a significant number of people who are of the opinion that the organisations which will prosper in future will be those who understand and work closest with their customers and suppliers. A natural conclusion from this is that leading companies will change the way they interact with others, to the point where they regard each other as partners. In true partnerships, they will share information and work co-operatively and collaboratively at all levels of operation. The early results from the carrier services trial are encouraging and they show how the business benefit from a deep e-business partnership can begin to be realised.

Acknowledgements

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References

- 1 Stockman S (Ed): 'Operational support systems', Special Themed Edition, BT Technol J, 15, No 1 (January 1997).
- 2 Mathieson G: 'Customer-handling intermediate server — an architecture-led project', BT Technol J, 15, No 1, pp 114—134 (January 1997).
- 3 Calladine J: 'BT Middleware — software as infrastructure', BT Technol J, 15, No 1, pp 135—146 (January 1997).
- 4 Dobson J and Martin M: 'The Ontology of Enterprises and Information Systems', Newcastle University (1997) — <http://hsb.baylor.edu/ramsower/ais.ac.97/papers/dobson.htm>

- 5 Revett M C, Knul M D T and Stephens L: 'Network computing', BT Technol J, 15, No 2, pp 172—178 (April 1997).
- 6 http://www.sybase.com/products/application_tools/
- 7 Briscoe R J: 'Distributed objects on the Web', BT Technol J, 15, No 2, pp 158—171 (April 1997).
- 8 Details of the Intranet Complete product are available from — <http://www.bt.com>
- 9 http://www.microsoft.com/products/prodref/599_ov.htm?RLD=21
- 10 http://www.microsoft.com/products/prodref/608_ov.htm?RLD=26
- 11 Rombach H D and Basili V R: 'The benefits of goal-oriented measurement', Proc 7th CSR Annual Conference on Software Reliability and Metrics (1990).



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Jonathan Mitchener joined BT Laboratories in 1988, having graduated from Hatfield Polytechnic with a first class BSc (Hons) in Computer Science. Initially working on architects for broadband services, he has, over 10 years, gained a wide range of experience in such areas as object-orientation, software reuse research, the Billing 90s programme, and WWW/Intranet developments to support Global Sales teams.

Since May 1996 he has led work on systems and software strategy issues across a variety of projects including current research into the future of e-business for new-wave telcos.



Andy Childerhouse graduated from the University of Hull with an honours degree in Management Systems. He joined BT in 1989 working initially on the NCDB. In 1991 he moved to system security, where he held a number of posts including BT Security Evaluation and Fraud Management. In 1996 he took up a post in Finance Billing Receivables where he was responsible for revenue assurance. This was followed by a term as a team leader within Interconnect Process and System Evolution.

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